

Shaoyuan Xie

Ph.D. Candidate, ICS, University of California, Irvine
Tel: (949)368-5560

Email: shaoyux@uci.edu
Homepage: <https://daniel-xsy.github.io>

EDUCATION

University of California, Irvine

Ph.D. in Computer Science, Advisor: Professor Qi Alfred Chen

Irvine, CA

Expected Sep 2027

Huazhong University of Science and Technology (HUST)

B.Eng. in Automation, GPA: 3.97 (Top 1st / 212 students)

Wuhan, CN

Jun 2023

WORKING EXPERIENCE

Waymo

Research Intern, AI Foundation | Mentor: Chen Song, Hao Xiang, Yang Liu

Mountain View, CA

Jun 2026 – Sep 2026

- Develop and research on Perception Foundation Model.

Mercedes-Benz Research & Development North America

Research Intern, AD Perception | Mentor: Thomas Monninger | Manager: Sihao Ding

San Jose, CA

Jun 2025 – Sep 2025

- Semantic distillation from LLM to BEV encoder for safer end-to-end driving.

Shanghai Artificial Intelligence Laboratory

Research Intern, OpenMMLab | Mentor: Wenwei Zhang | Manager: Kai Chen

Shanghai, CN

Jun 2023 – Aug 2023

- [InternLM](#) language models post training.

TOP-TIER PUBLICATIONS

Summary

2 (1 first-author) in top-tier machine learning conferences (NeurIPS, ICLR)

3 (3 first-author) in top-tier computer vision conferences and journals (TPAMI, ICCV, ECCV)

3 (1 first-author) in top-tier security conferences (NDSS)

2 (1 first-author) in top-tier robotics conferences (CoRL, IROS)

- **SafeDriveVLA: Navigation-Conditioned World Model Dreaming for Conflict-Aware End-to-End Autonomous Driving**

Shaoyuan Xie*, Zihan Zhang*, Jingxuan Wang, Jiashu Qu, Xiaoqing Liang, Lingdong Kong, Junchi Lu, Henrik I Christensen, Qi Alfred Chen

CoRL, 2026 (acceptance rate 32.8% = 687/2094)

- **BEVLM: Distilling Semantic Knowledge from LLMs into Bird's-Eye View Representations**

Thomas Monninger*, Shaoyuan Xie* (*order determined alphabetically), Qi Alfred Chen, Sihao Ding

ECCV, 2026 (acceptance rate 27.5% = 2883/10473)

- **FlyTrap: Physical Distance-Pulling Attack Towards Camera-based Autonomous Target Tracking Systems**

Shaoyuan Xie, Mohamad Fakih, Junchi Lu, Fayzah Alshammari, Ningfei Wang, Takami Sato, Halima Bouzidi, Mohammad Al Faruque, Qi Alfred Chen

NDSS, 2026 (acceptance rate 17.9% = 265/1481)

- **Are VLMs Ready for Autonomous Driving? An Empirical Study from Reliability, Data, and Metric Perspectives**

Shaoyuan Xie, Lingdong Kong, Yuhao Dong, Chonghao Sima, Wenwei Zhang, Qi Alfred Chen, Ziwei Liu, Liang Pan

ICCV, 2025 (acceptance rate 24.0% = 2699/11239)

- **Benchmarking and Improving Bird's Eye View Perception Robustness in Autonomous Driving**

Shaoyuan Xie, Lingdong Kong, Wenwei Zhang, Jiawei Ren, Liang Pan, Kai Chen, Ziwei Liu

TPAMI, 2025 (Impact Factor = 20.4)

- **Can We Trust Embodied Agents? Exploring Backdoor Attacks against Embodied LLM-Based Decision-Making Systems**
Ruo Chen Jiao*, Shaoyuan Xie* (*co-first author), Justin Yue, Takami Sato, Lixu Wang, Yixuan Wang, Qi Alfred Chen, Qi Zhu
ICLR, 2025 (acceptance rate 32.1% = 3689/11500)
- **The Heat is On: Understanding and Mitigating Vulnerabilities of Thermal Image Perception in Autonomous Systems**
Hrushikesh Varma, Shaoyuan Xie, Michael Clifford, Qi Alfred Chen, Takeshi Sugawara, Sara Rampazzi
NDSS, 2026 (acceptance rate 17.9% = 265/1481)
- **Revisiting Physical-World Adversarial Attack on Traffic Sign Recognition: A Commercial Systems Perspective**
Ningfei Wang, Shaoyuan Xie, Takami Sato, Yunpeng Luo, Kaidi Xu, Qi Alfred Chen
NDSS, 2025 (acceptance rate 15.4% = 211/1369)
- **RoboDepth: Robust Out-of-Distribution Depth Estimation under Corruptions**
Lingdong Kong, Shaoyuan Xie, Hanjiang Hu, Lai Xing Ng, Benoit R. Cottreau, Wei Tsang Ooi
NeurIPS, 2023 (acceptance rate 26.1% = 3218/12343)
- **Adversarial Spatial Transform Marker: A Black-Box Pose Estimation Attack for Autonomous UAV Landing**
Junchi Lu, Fayzah Alshammari, Shaoyuan Xie, Xiaoqing Liang, Qi Alfred Chen
IROS, 2026

OTHER PEER-REVIEWED PUBLICATIONS

- **"Prompter Says": A Linguistic Approach to Understanding and Detecting Jailbreak Attacks Against Large-Language Models**
Dylan Lee, Shaoyuan Xie, Shagoto Rahman, Kenneth Pat, David Lee, Qi Alfred Chen
ACM CCS Workshop - LAMPS, 2024
- **On the Adversarial Robustness of Camera-based 3D Object Detection**
Shaoyuan Xie, Zichao Li, Zeyu Wang, Cihang Xie
TMLR, 2024

RESEARCH IMPACT

- Email acknowledgements from DJI on autonomous tracking vulnerabilities
- Email acknowledgements from automotive manufacturer on traffic sign recognition vulnerabilities (One company among Tesla, Toyota, Nissan, Mazda, and Hyundai; no direct revelation of the company identity due to the anonymity agreement achieved during the responsible vulnerability disclosure process)

AWARDS AND HONORS

- Best Demo, 3rd USENIX Symposium on Vehicle Security and Privacy (2025, *selected for best demo rate 18.2% = 2/11*)
- Honours Bachelor's Degree (2023, *top 1% in univ.*)
- Outstanding Undergraduate Student (2021, *top 1% in univ.*)
- National Scholarship, Ministry of Education of China (2021, *top 0.2% nation-wide*)
- National Scholarship, Ministry of Education of China (2020, *top 0.2% nation-wide*)

ACADEMIC SERVICES

- *Competition Organizer*: IROS RoboSense (2025), ICRA RoboDrive (2024), ICRA RoboDepth (2023)
- *PC member*: ACM CCS Artifact Evaluation (2024, 2025, 2026), NDSS Artifact Evaluation (2025)
- *Reviewer*: CVPR (2023, 2024, 2025, 2026), ICCV (2025), ECCV (2026), BMVC (2026), ICLR (2026), CoRL (2026), IROS (2026), IJCV, RA-L, TDSC; TMM, TMLR
- *External Reviewer*: USENIX Security (2024, 2025), IEEE S&P (2024), ACM CCS (2024)